

شارك | SHARE-K

AI-Powered Open Source Collaboration Hub

Project Pitch

ICT Training Institution (ITI) • Capstone — Full Stack Development

Project Title	Share-k — AI-Powered Open Source Collaboration Hub
Team Name	Share-k Dev Team
Target Industry	Developer Tools & Productivity / Open Source / EdTech
Team Members	6-Member Full Stack Team
Version	MVP — January 2026

Connecting Developers. Validating Skills. Powering Open Source.

1 THE CORE PROBLEM (THE "WHY")

Open-source development powers the modern technology industry, yet it suffers from critical collaboration inefficiencies that harm both project maintainers and aspiring developers.

Who Suffers Most:

- Project owners post contribution tasks but receive applications from developers who lack the required skills, wasting hours reviewing unsuitable candidates.
- Developers struggle to discover real projects that match their actual skill level, stifling career growth and portfolio development.
- There is no trusted, verified reputation system for contributor quality outside of raw GitHub metrics.
- Developers who are rejected receive no actionable guidance on how to improve or become eligible for future tasks.

The Pain Point	Who Experiences It
Unqualified applications flood project owners	Open-source maintainers & team leads
No structured contributor discovery	Junior & mid-level developers
No verified reputation outside GitHub stars	Both owners and contributors
Zero feedback on rejected applications	Developers seeking career growth

2 THE SOLUTION & VALUE PROPOSITION (THE "WHAT")

Share-k is an AI-powered open-source collaboration hub that eliminates these bottlenecks with a full-stack, AI-first platform. It uses LLM-driven skill profiling, RAG-powered contributor matching, and a multi-agent orchestration system to intelligently connect project owners with the right contributors — before a single application is sent.

The Single Most Important Benefit:

AI validates every contributor application BEFORE it reaches the project owner — eliminating wasted review time while creating a structured growth pathway for developers.

Four AI-Driven Solution Pillars:

Pillar	Capability	Technology
Smart Skill Profiling	Automatically analyzes GitHub repos and generates verified skill profiles	LLM + GitHub API + RAG

Intelligent Matching	Matches contributors to tasks based on actual technical fit	Agentic AI + Vector Search
Validation Gating	Blocks unqualified applications before they reach project owners	AI Skill Validation Agent
Growth Guidance	Provides personalized learning paths to close skill gaps	RAG + LLM Generation

3 THE AI TRINITY STRATEGY (THE "HOW")

Share-k is built on three deeply integrated AI pillars that create a defensible technical moat and a superior developer experience:

□ LLM — Intelligence Layer (Examples)

Foundation Model: OpenAI GPT-4o (primary) + Anthropic Claude 3 Haiku (fallback)

Primary Role: GPT-4o serves as the reasoning backbone of every AI agent — performing skill extraction from GitHub code, chain-of-thought eligibility validation, structured JSON skill profiling, and natural language guidance generation. Advanced API features used include function calling (tool use) in the Skill Validation Agent and token-by-token streaming for gap guidance reports. Claude 3 Haiku handles lower-stakes tasks such as notifications and summaries for cost efficiency.

□ RAG — Knowledge Layer

Data Sources the AI Reads:

GitHub README files, code files (.py, .js, .ts, .go), commit messages, and repo descriptions form the primary knowledge base for skill extraction. Contributor profiles indexed in Pinecone (vector DB) power contributor-to-task matching. A curated learning resource index — documentation summaries and tutorials — fuels personalized skill gap guidance. Published project metadata enables semantic project discovery.

Advanced RAG Implementation: Hybrid retrieval (semantic vector search + BM25 keyword) with Reciprocal Rank Fusion, cross-encoder reranking (top-20 → top-5), and Query Transformation — decomposing task descriptions into targeted sub-queries for maximum retrieval recall.

□ Agents — Action Layer

Four Specialized Agents in a LangGraph Sequential Pipeline:

1. GitHub Ingestion & Skill Profiling Agent — On user registration, autonomously fetches all GitHub repos, analyzes code with ReAct reasoning, and generates a structured JSON skill profile [{skill, level, confidence, source_repo}].
2. Skill Validation Agent — When a contributor clicks 'Apply', autonomously compares their verified profile against task requirements using chain-of-thought reasoning, then issues an ELIGIBLE/INELIGIBLE decision with confidence score and justification — blocking unqualified applications before they reach the project owner.
3. Skill Gap Guidance Agent (Gold Tier) — On rejection, autonomously retrieves curated learning resources and generates a personalized roadmap: {missing_skills, recommended_resources, practice_projects, estimated_weeks}.

4. Contributor Matching Agent (Owner Gold/Silver) — When a new task is published, autonomously ranks the top-N best-fit contributors by skill alignment and notifies them proactively.

4 MARKET & COMPETITION

Primary Users:

Project Owners	Contributors
Open-source developers and teams maintaining public repositories who need qualified contributors but lack a structured way to find and vet them.	Junior to mid-level developers seeking real-world experience, portfolio projects, and a verified technical reputation beyond raw GitHub metrics.

Competitive Advantage — Why Share-k Wins:

Dimension	Share-k	GitHub Issues / Existing Tools
Contributor Matching	✓ AI-powered skill validation + semantic matching	✗ Manual review of all applicants
Skill Verification	✓ AI-extracted from actual code, not self-reported	✗ Self-declared skills only
Rejection Experience	✓ Gold tier: AI learning roadmap generated instantly	✗ No guidance provided
Reputation System	✓ Multi-metric, contribution-verified public profile	✗ Star counts only
Application Quality	✓ Only qualified applicants reach owners	✗ All applications reach owners unfiltered

5 EXECUTION PLAN (HIGH LEVEL)

Core Technology Stack:

Frontend	React (Next.js) — SSR, Tailwind CSS, RTL/Arabic support, WCAG 2.1 Level AA
Backend	Python (FastAPI) — async, OpenAPI auto-docs, Celery + Redis for background agent tasks
AI Framework	LangChain + LangGraph — stateful multi-agent pipeline orchestration
Vector Database	Pinecone (serverless) — HNSW indexing, separate namespaces per data type
Primary Database	PostgreSQL (AWS RDS) — skill profiles, application history, reputation events
Cloud Provider	AWS — ECS Fargate (backend), Amplify (frontend), S3 (file storage)

Observability	Langfuse (LLM traces) + CloudWatch (infra) + Sentry (error tracking)
CI/CD	GitHub Actions — pytest + Vitest + Docker build + ECR push + ECS deploy

MVP Timeline: 4 Weeks — 6-Member Team

Week 1	User auth, GitHub OAuth integration, Skill Profiling Agent, PostgreSQL + Pinecone setup
Week 2	Project publishing flow, Skill Validation Agent, AI-gated application pipeline
Week 3	Skill Gap Guidance Agent, Contributor Matching Agent, subscription tiers, reputation system
Week 4	Observability (Langfuse), E2E testing (Playwright), CI/CD pipeline, deployment to AWS

Risk Factor — Biggest Technical Challenges:

Risk	Mitigation Strategy
Skill extraction accuracy on sparse repos	Admin human review gate before profile activation; accuracy degrades gracefully and is flagged for manual assessment
GitHub API rate limits during bulk ingestion	Dynamic replanning in the Ingestion Agent; queues retries for failed repos; OAuth increases rate limits significantly
LLM hallucination in skill profiles	Faithfulness metric (>90% target) enforced via Langfuse; citation bubbles link each skill to its source repo as ground-truth evidence
LLM API cost at scale	Model tiering (GPT-4o only for high-value tasks), embedding caching, query caching, token budgets, batch processing for matching jobs
Prompt injection in task descriptions	Llama Guard screening on all user inputs; indirect injection defense in the GitHub README ingestion pipeline; input sanitization middleware

*"Without the AI, Share-k is just another job board.
With it, it becomes a trusted infrastructure layer for the open-source development economy."*